

1. Square of a directed graph algorithm

Create $G_{\text{square}}(A, n)$:

$A2 \leftarrow n \times n$ matrix of 0s

for $i = 1$ to n :

for $j = 1$ to n :

// is there

*check if there's a k with
 $i \rightarrow k$ and $k \rightarrow j$

for $k = 1$ to n :

if $A[i][k] == 1$ and $A[k][j] == 1$:

$A2[i][j] \leftarrow 1$

break

*done for this (i, j)

return $A2$;

Running time = $\Theta(n^3)$ because
of 3 nested loops.

2.

a) Cycle_DFS(G)

for each vertex u : $color[u] \leftarrow \text{White}$, $parent[u] \leftarrow \text{NIL}$

for each vertex u :

if $color[u] = \text{White}$ and $\text{DFS_visit}(u) = \text{true}$: return "cyclic"
return "acyclic"

DFS_visit(u):

$color[u] \leftarrow \text{Gray}$

for each v in $Adj[u]$:

if $color[v] = \text{White}$: $parent[v] \leftarrow u$

if $\text{DFS_visit}(v) = \text{True}$: return true

else if $v \neq parent[u]$:

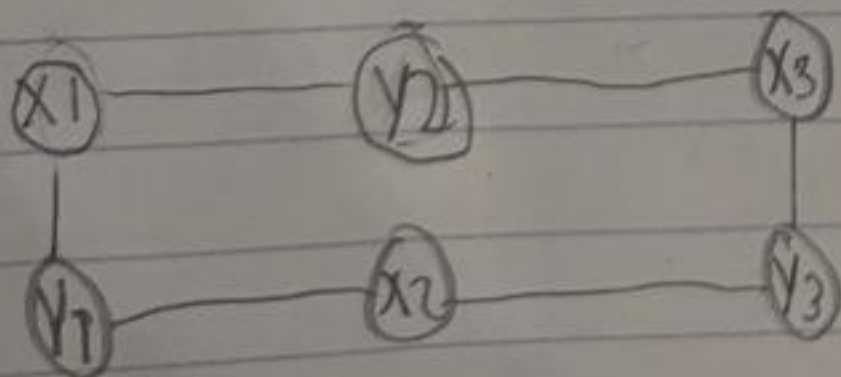
return true

$color[u] \leftarrow \text{black}$

return false

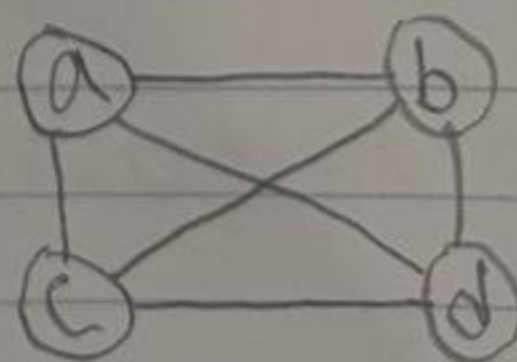
(i) (6 nodes) adjacency list

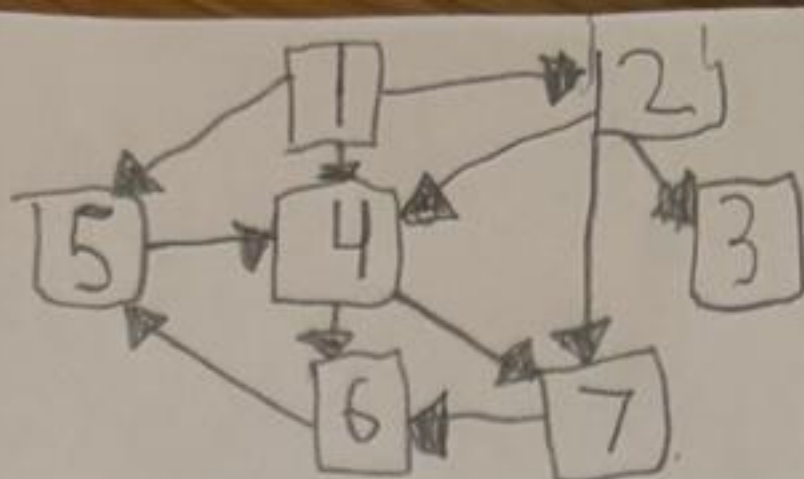
(i)



(ii) (4 nodes) Adjacency Matrix

(ii)





b)

c)

1: 2, 4, 5
 2: 3, 4, 7
 3: 4
 4: 6, 7
 5: 4
 6: 5
 7: 6

visit

Order: 1, 2, 4, 5, 3, 7, 6

3. Does an undirected graph exist w 5 vertices each of degree 3?

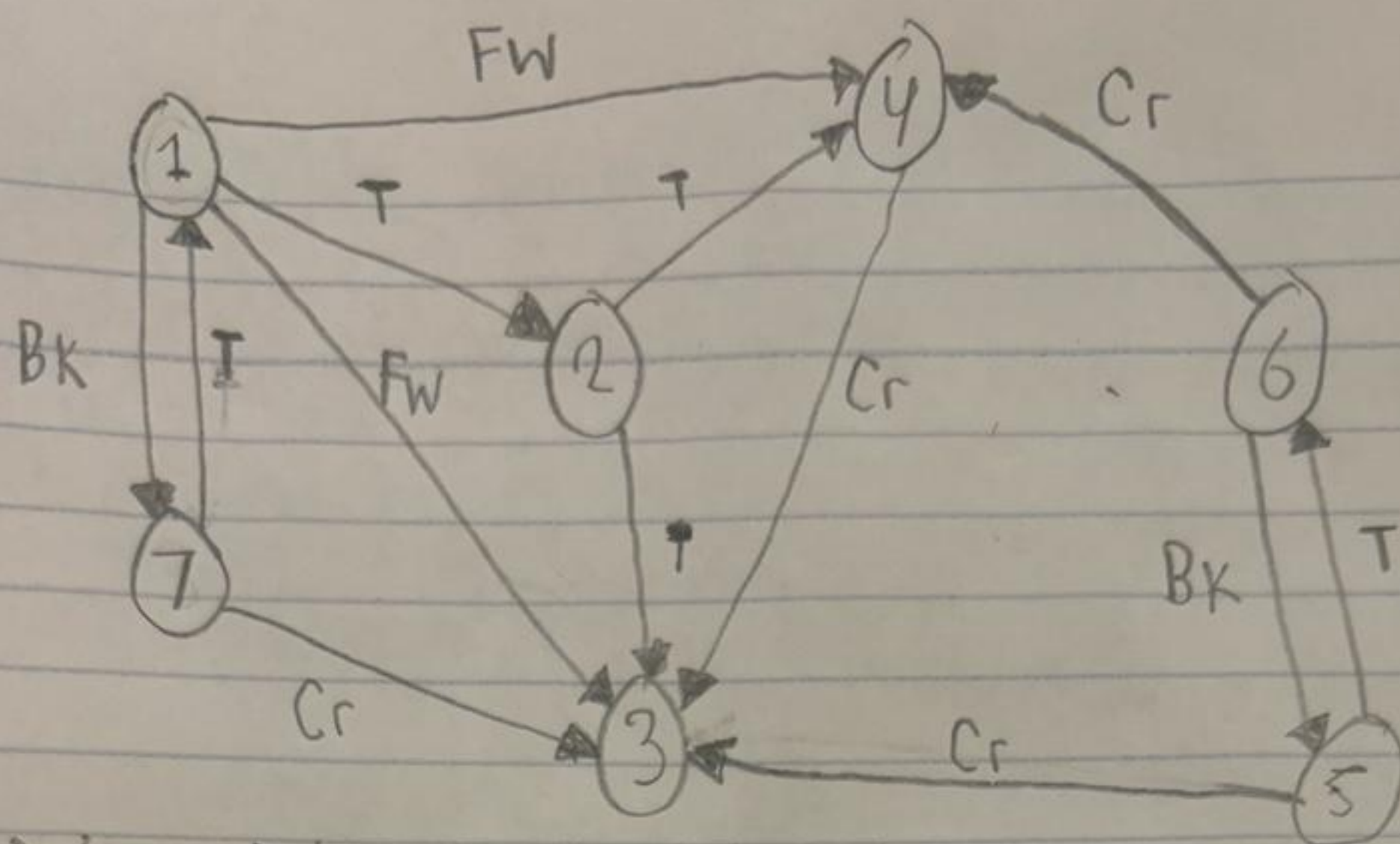
No.

Assume Undirected graph G with $n = 5$ vertices, each of deg 3. Sum of degrees is:

$$\sum \deg(v) = 5 \cdot 3 = 15$$

by the handshaking lemma it must be an even # but 15 is odd which contradicts and shows such a graph can't exist.

4.



T: (1 → 2), (2 → 3), (2 → 4), (1 → 7), (5 → 6)

Bk: (7 → 1), (6 → 5)

Fw: (1 → 3), (1 → 4)

Cr: (4 → 3), (7 → 3), (5 → 3), (6 → 4)

5. Discovery/finishing times

V:	d[V]	f[V]
1:	1	12
2:	2	11
3:	3	4
4:	5	8
8:	6	7
7:	9	10
6:	13	16
5:	14	15

Discovery order (nodes found): 1, 2, 3, 4, 8, 7, 6, 5

Finishing order (by completion): 3, 8, 4, 7, 2, 1, 5, 6

Topological order (reverse finishing): 6, 5, 1, 2, 7, 4, 8, 3

6. Bipartite graph algorithm

For all $v \in V$ set $\text{color}[v] = \text{Uncolored}$

For each vertex s :

if $\text{color}[s] == \text{uncolored}$ then

$\text{color}[s] = 0$, push s in queue Q .

~~enqueue~~

While Q is not empty:

$u = \text{pop } Q$

for each neighbor w of u :

if $\text{color}[w] = \text{Uncolored}$:

$\text{color}[w] = 1 - \text{color}[u]$; Push w

else if $\text{color}[w] = \text{color}[u]$:

return "Not bipartite" (odd cycle exists)

if finished with no conflict, return "bipartite" with
 $V_1 = \{v \mid \text{color}[v] = 0\}$, $V_2 = \{v \mid \text{color}[v] = 1\}$

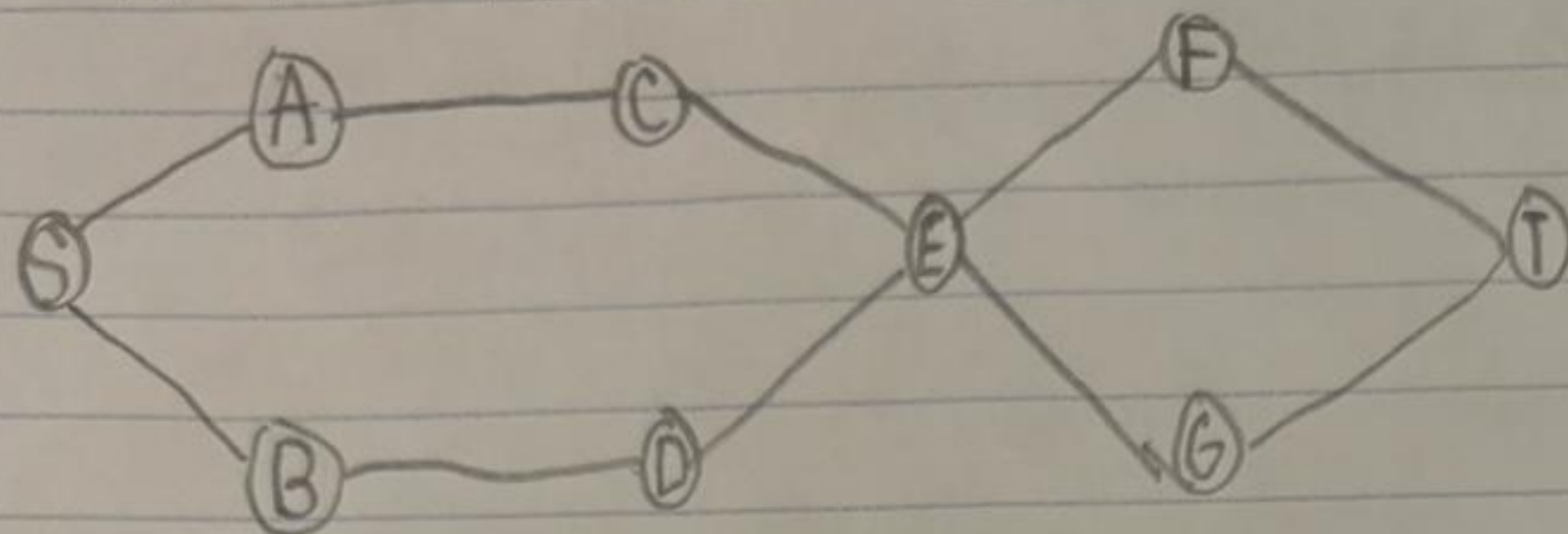
BFS alternates colors at each level

If a conflict (same color edge) appears, it means an odd cycle exists

If no conflict occurs, all edges connect opposite colors, so the graph is bipartite

Time complexity: $\Theta(|V| + |E|)$ since each vertex and edge is checked once.

7. Graph for maze:



(Undirected)

S: A, B
A: S, C
B: S, D
C: A, E
D: B, E
E: C, D, F, G
F: E, T
G: E, T
T: F, G

I would use BFS So we can find the shortest path as DFS might find a longer path than the most optimal, so BFS.

8. $x = \#$ of \$250 units, $y = \#$ of \$400 units

Maximize $P = 45x + 50y$

Such that

$$x + y \leq 250$$

$$250x + 450y \leq 70,000$$

$$x, y \geq 0$$

then check corner points

$$(250, 0), (0, 175)$$

and the intersection
of the two lines

intersection:

$$x = 250 - y$$

$$250(250 - y) + 400y = 70,000$$

$$62,500 - 250y + 400y = 70,000 \rightarrow 150y = 7,500$$

$$y = 50, x = 200$$

Profits:

(x, y)	P
$(250, 0)$	$45(250) = 11,250$
$(0, 175)$	$50(175) = 8,750$
$(200, 50)$	$45(200) + 50(50) = 11,500$

$$x = 200 \quad y = 50 \quad P_{\max} = \$11,500$$

Simplex Algorithm for Question #9

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Solving using Simplex Method...

Problem: Maximize $P = 20x_1 + 10x_2 + 15x_3$

Initial table:

	x1	x2	x3	s1	s2	s3	s4	RHS
s1	3	2	5	1	0	0	0	55
s2	2	1	1	0	1	0	0	26
s3	1	1	3	0	0	1	0	30
s4	5	2	4	0	0	0	1	57
P	-20	-10	-15	0	0	0	0	0

--- Iteration 1 ---

Entering variable: x1

Ratio test:

Row 0: $55/3 = 18.3333$

Row 1: $26/2 = 13$

Row 2: $30/1 = 30$

Row 3: $57/5 = 11.4$

Leaving variable: s4

Pivot element: 5

Making pivot element 1 by dividing row 3 by 5

Making row 0 element 0 using factor 3

Making row 1 element 0 using factor 2

Making row 2 element 0 using factor 1

Making row 4 element 0 using factor -20

After pivot operations:

	x1	x2	x3	s1	s2	s3	s4	RHS
s1	0	0.8	2.6	1	0	0	-0.6	20.8
s2	0	0.2	-0.6	0	1	0	-0.4	3.2
s3	0	0.6	2.2	0	0	1	-0.2	18.6
s4	1	0.4	0.8	0	0	0	0.2	11.4
P	0	-2	1	0	0	0	4	228

--- Iteration 2 ---

Entering variable: x2

Ratio test:

Row 0: $20.8/0.8 = 26$

Row 1: $3.2/0.2 = 16$

Row 2: $18.6/0.6 = 31$

Row 3: $11.4/0.4 = 28.5$

Leaving variable: s2

Pivot element: 0.2

Making pivot element 1 by dividing row 1 by 0.2

Making row 0 element 0 using factor 0.8

Making row 2 element 0 using factor 0.6

Making row 3 element 0 using factor 0.4

Making row 4 element 0 using factor -2

After pivot operations:

	x1	x2	x3	s1	s2	s3	s4	RHS
s1	0	0	5	1	-4	0	1	8
s2	0	1	-3	0	5	0	-2	16
S3	0	0	4	0	-3	1	1	9
s4	1	0	2	0	-2	0	1	5
P	0	0	-5	0	10	0	1.97215e-31	260

--- Iteration 3 ---

Entering variable: x3

Ratio test:

Row 0: $8/5 = 1.6$

Row 2: $9/4 = 2.25$

Row 3: $5/2 = 2.5$

Leaving variable: s1

Pivot element: 5

Making pivot element 1 by dividing row 0 by 5

Making row 1 element 0 using factor -3

Making row 2 element 0 using factor 4

Making row 3 element 0 using factor 2

Making row 4 element 0 using factor -5

After pivot operations:

	x1	x2	x3	s1	s2	s3	s4	RHS
s1	0	0	1	0.2	-0.8	0	0.2	1.6
S2	0	1	0	0.6	2.6	0	-1.4	20.8
s3	0	0	0	-0.8	0.2	1	0.2	2.6
s4	1	0	0	-0.4	-0.4	0	0.6	1.8
P	0	0	0	1	6	0	1	268

=== OPTIMAL SOLUTION ===

x1 = 1.8

x2 = 20.8

x3 = 1.6

Maximum P = 268

Running time: 0.267 milliseconds

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